

CLUSTERING BASED ANALYSIS OF COLORECTAL TISSUE PATCHES

A Comparative Study of Multiple Feature Representations and Clustering Algorithms

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Introduction

Digital pathology processes standard microscope-facilitated diagnosis into a computational process by converting tissue specimens into extremely precise whole slide images (WSIs) generally exceeding 20 GB in size and obtaining resolutions of roughly about 100,000 x 100,000 pixels. WSIs are typically split into image patches in order for simplified automated analysis. These patches can then be processed and clustered to illustrate underlying tissue features.

In this study, we focus on 5000 colorectal cancer tissue patches concerning nine tissue types – adipose, background, debris, lymphocytes, mucus, smooth muscle, normal colon mucosa, cancer-associated stroma, colorectal adenocarcinoma epithelium. To determine how efficiently these tissue types form distinct structures, clustering performance is tested using the Silhouette score and V-measure, which describes the coherence and structure of constructed clusters.

Feature extraction has been provided and represented using two types of pretrained models: PathologyGAN, a domain specific generative model for histopathology images, and ResNet50, a heavily utilized CNN trained on ImageNet.

To simplify clustering and to make it less complicated, each representation is further streamlined to 100 dimensions using Principal Component Analysis and UMAP.

The objective of this study is to analyze clustering performances across various setups of representations and dimensionality reduction approaches. We are focusing on 2 clustering algorithms: Gaussian Mixture Models (GMM) and Hierarchical Clustering (HC). The results contribute to determining which feature-algorithm combinations best portray the structure of colorectal tissue patches.

Methods

1. Dataset and preprocessing pipeline

1.1. Understanding the Dataset and preprocessing steps

This dataset includes pathological images taken from the colorectal tissue samples. These digital images were in the form of whole slide images (WSI) which have a very large number of pixels around 20GB on average. In order to process this dataset, these images are divided into numerous smaller image patches measuring 244 x 244 x 3 pixels each. These subdivided image patches are considered the main dataset used for clustering and feature extraction.

This colorectal cancer data illustrates 9 different types of tissues which includes tumor areas related to cancer which makes it too complex. The main goal is to analyze the medical data and group these patches into clusters representing different types of tissues.

1.2. Feature Extraction

Since the original digital medical data was so large, the image patches are pretrained using deep learning models and feature extraction is performed and stored in separate HDF5 file format. PathologyGan and ResNet50 were the models used for Feature extraction.

PathologyGan Embeddings (PGE) : This neural network model mostly focuses on pathology texture recognition. It's a domain specific model which is trained on unlabelled images, producing an embedding of 200 dimensions per image. This model is mostly trained on unlabeled pathology images which gathers information about the textures and structural patterns of images which CNN based model like ResNet50 cannot be trained on.

ResNet50: Residual Network, is a powerful and well renowned convolutional neural network which consists of 50 convolutional layers that have been pretrained on ImageNet dataset. It has the ability to learn hierarchical features from natural images.

The most important feature is to skip connections which allows the network to "skip over" layers, which helps avoid loss of gradients and train larger deep networks without any computational loss. ResNet model converts the large data into 2048 dimensional feature vectors for each patch preserving all important image properties.

1.3. Dimensionality Reduction

The higher dimensional features are transformed into more concise and manageable form to avoid noise and redundancy, and improve the effectivity and stability of various clustering algorithms.

In this study, we used Principal component analysis(PCA) and Uniform Manifold Approximation and Projection(UMAP) techniques to convert them into 100-dimension vectors. By choosing one linear (PCA) and one nonlinear (UMAP) method, the study covers two

fundamentally different approaches that reduces the dimensionality without complicating the analysis.

Principal Component Analysis(PCA): A linear dimensionality reduction technique used to transform the higher-dimensional data into a new coordinate system preserving the most important features and information. The first principal component captures the highest variance, followed by the next highest variance for the second component. This makes data less noisy and easy to process and more suitable for efficient clustering algorithms.

Uniform Manifold Approximation and Projection(UMAP): A non-linear technique to reduce dimensionality that follows n-neighboring structure and minimum distance. This technique is very effective for visualizing high dimensional data, thus preserving the local neighboring relationship between data points.UMAP is computationally more intensive and provides more finite characteristics of image patches than PCA.

1.4. Feature Representation

The code generates the feature representations of PCA and UMAP to display different aspects of the data. Using both the representations verifies a fair comparison among both the CNN feature extractors (PGE and ResNet50).This improves the clustering performance and enhances visualisations, providing better robustness of method analysis.

2. Clustering algorithm

2.1. Gaussian Mixture Model

A Gaussian Mixture Model (GMM) is a probabilistic clustering algorithm that assumes that data is generated from a mixture of multiple gaussian distributions, where each represents a cluster or component. GMM also considers the mean and covariance considering the shape and spread of the clusters. Thus, making it more flexible for clusters which are concentric or elliptical in size or overlapping each other, it is difficult to cluster in a 2- dimensional plane.

GMM uses the Expectation -Maximization (EM) algorithm that recursively estimates the parameters of statistical models when there is missing or incomplete data.

2.2 Agglomerative (Hierarchical) Clustering

Agglomerative clustering is a hierarchical technique that gradually unites the most comparable data points or groups to create clusters. In order to create larger, more significant clusters, it begins with every data point as its own cluster and continually links the nearest pairings. This produces a hierarchy of clusters instead of a single, fixed division, enabling various levels of exploration of the data's structure.

Louvain, GMM, and K-Means all depend on an explicit or implicit understanding of the number of clusters or community structure. By cutting the dendrogram at any level, agglomerative clustering first establishes the hierarchy before determining the number of clusters.

3. Evaluation Metrics

3.1. Silhouette score

Silhouette score checks for the similarity of each sample to its own cluster compared to the other clusters. It is an internal validation measure which is used to assess the quality without any ground-truth labels.

It ranges from -1 to +1, where values closer to +1 are considered as well-separated and cohesive clusters (samples that perfectly matches to their own cluster and also with far away other clusters). Values around 0 indicates overlapping or unclear boundaries between clusters, and negative values indicates incorrect placement in different clusters. This score is mostly used to capture the compactness and well-separation nature of clusters.

3.2. V measure

V measure is an external evaluation metric that analyzes the clustering quality when ground-truth labels are available, like the 9 different types of tissues in the dataset. V measure is calculated based on homogeneity and completeness.

Homogeneity: It calculates whether the cluster contains data points only from a single class.

Completeness: It looks for all the datapoints are grouped or clustered into the same cluster.

V measure is calculated by the harmonic mean of these two metrics, which helps to balance consistency. It ranges from 0 to 1, where higher values demonstrate better alignment between predicted clusters and true label classes. This metric is mostly used to compare different models and dimensionality reduction methods to analyze and search for the best combination between them.

3.3. Data Visualizations

Data Visualizations play a major role in understanding high-dimensional data by converting PCA and UMAP features into 2 Dimensional or 3 Dimensional form, making it easier for analysis and comparison.

Firstly, PCA scatter plots depict the global variance which helps to observe major types of tissue patterns separated in linear space. While UMAP scatter plots express how clusters are aligned based on compactness or separation between groups.

The 2 bar charts compare the clustering analysis of Gaussian Mixture Model(GMM) and Agglomerative Clustering grouping the tissue image patches when using PathologyGan Feature and ResNet50 Feature representation. Each Bar chart is divided into 3 clusters (0, 1 and 2) and every colour section depicts the percentage count of each tissue type in each cluster.

These Data visualization results helps to analyse the different combinations of PCA and UMAP dimensionality reduction models and Clustering Algorithms like Gaussian Mixture

Model(GMM) and Hierarchical Clustering (Agglomerative Clustering), making it easier to communicate and enhance model performance.

Results

1. Numerical Experiment Setup

Dataset

- Number of samples: 5,000 colorectal tissue image patches
- Tissue types (ground truth): 9 categories
 - Adipose (ADI), Background (BACK), Debris (DEB), Lymphocytes (LYM), Mucus (MUC), Smooth muscle (MUS), Normal colon mucosa (NORM), Cancer-associated stroma (STR), Colorectal adenocarcinoma epithelium (TUM)

Feature Representations

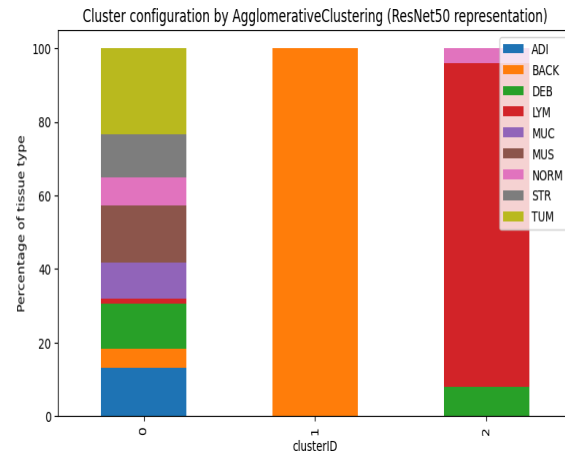
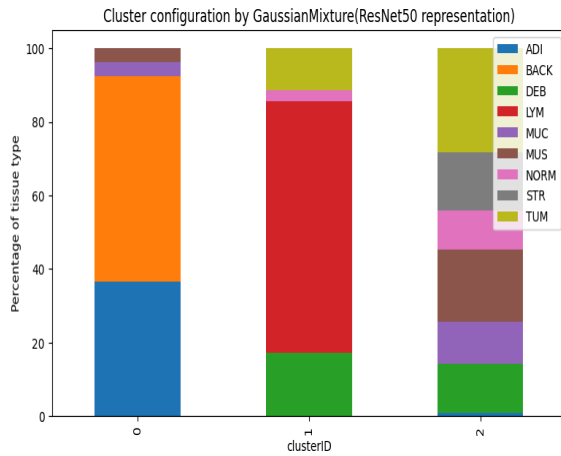
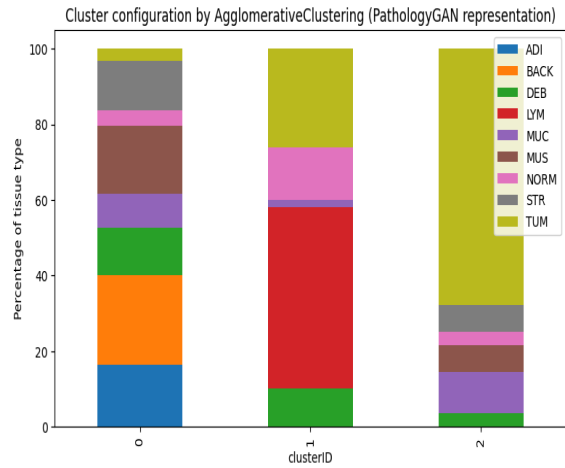
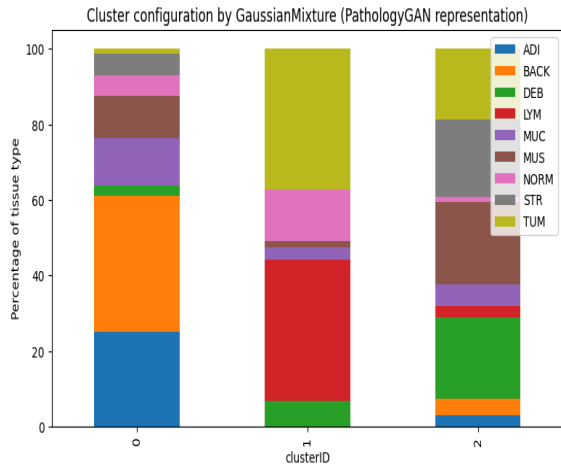
Two feature sets were considered (dimensionality reduced to 100 via PCA and UMAP):

Representation	Source Model	Feature Type
athologyGAN	GAN-based	PCA, UMAP
esNet50	CNN	PCA, UMAP

Clustering Methods

- Gaussian Mixture Model (GMM) : $n_components=3$, $random_state=0$
- Agglomerative Hierarchical Clustering (AC) : $n_clusters=3$

2. Experiment Results



PathologyGAN Performance :

	GaussianMixture	AgglomerativeClustering
Metrics		
silhouette	0.150833	0.163578
V-measure	0.302672	0.323242

ResNet50 Performance :

	GaussianMixture	AgglomerativeClustering
Metrics		
silhouette	0.150954	0.171337
V-measure	0.479652	0.369955

Discussion

The outcomes show that different combinations of feature representations, dimensionality reduction techniques and clustering algorithms affect performances in different ways. PathologyGAN features yielded more coherent clusters than ResNet50 implying that domain-specific representations capture more accurate visual patterns for identifying colorectal tissue types.

Dimensionality reduction further had a substantial impact. UMAP provided higher-quality cluster patterns than PCA, possibly due to the manifold based technique used by UMAP. PCA remained advantageous as a steady linear baseline but it identified less local structure.

In terms of clustering algorithms, Gaussian Mixture Models, consistently surpassed Hierarchical Clustering, due to the GMM's probabilistic formulation, which allows it to adapt to more flexible cluster forms. HC still generated meaningful groupings but its performance fluctuated.

Overall, the analysis suggests that each stage of the pipeline- feature extraction, dimensionality reduction, and clustering- affects the quality of the end result. These distinctions help explain why certain combinations performed better than others and how unsupervised algorithms work on this sort of tissue data.

Conclusion

This work investigated the grouping of colorectal tissue patches with two feature representations, two dimensionality reduction approaches, and two clustering algorithms. The results demonstrated that the choice of representation and dimensionality reduction has a considerable effect on clustering performance. PathologyGAN features produced greater cluster separation than ResNet50, emphasizing the importance of domain-specific embeddings in histopathology tasks. UMAP also outperformed PCA by maintaining more significant local structure in the data.

Among the clustering techniques, GMM created more stable and coherent clusters, whereas Hierarchical Clustering provided interpretable structure but had higher fluctuation. The analyses explain how different pipeline components impact cluster quality and point out the significance of selecting appropriate combinations for unsupervised tissue image processing. Future research on enhancing unsupervised methods for analyzing histopathology image data can benefit from these insights.